# ANN ASSIGNMENT

Digit Recognition with LeNet-5 on MNIST

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GitHub repository: ann-assignment

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## 1 Dataset

The dataset used in this project is MNIST, a standard benchmark for handwritten digit recognition. The program loads it through `torchvision.datasets.MNIST` with separate training and test splits. In the code, the dataset contains 70,000 images in total:

- 60,000 training images
- 10,000 test images

Each image belongs to one of 10 classes, representing digits 0 through 9. Every sample is a 28×28 grayscale image, which the program converts into a 1×28×28 tensor for the CNN. If flattened, each image contains 784 pixel values.

Table 1: MNIST dataset summary

| Property | Value |
|---|---|
| Total images | 70,000 |
| Training images | 60,000 |
| Test images | 10,000 |
| Number of classes | 10 |
| Image size | 28×28 grayscale |
| Program input tensor | 1×28×28 |
| Flattened pixel values | 784 |

## 2 LeNet-5 Model

LeNet-5 is one of the earliest and most influential convolutional neural network architectures for digit classification. In this project, it is adapted for MNIST by using one grayscale input channel and a padded first convolution so the spatial size remains 28×28 before pooling.

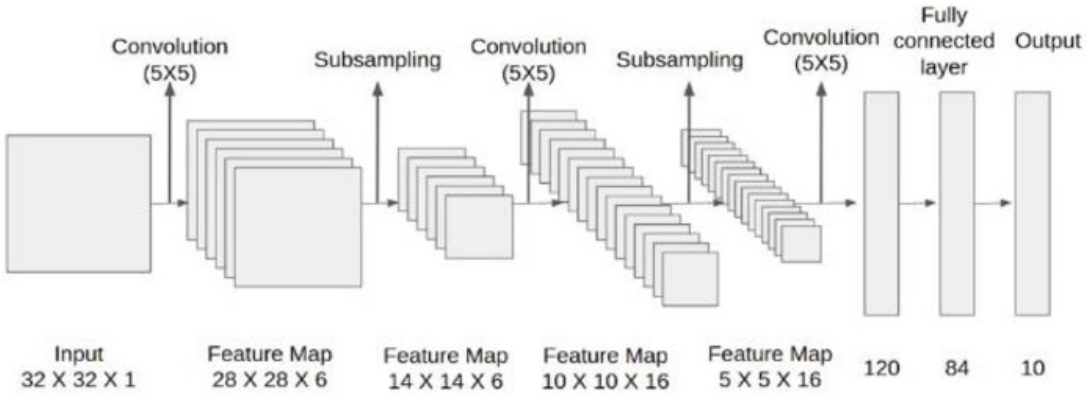


Figure 1: LeNet-5 CNN architecture reference used for the project report.

Table 2: Model architecture used in the project

| Layer | Configuration / Output | Parameters |
|---|---|---|
| Input | 1×28×28 grayscale image tensor | 0 |
| Conv1 | 6 filters, 5×5 kernel, padding 2, Tanh; output 6×28×28 | 156 |
| Pool1 | Average pooling, 2×2, stride 2; output 6×14×14 | 0 |
| Conv2 | 16 filters, 5×5 kernel, Tanh; output 16×10×10 | 2,416 |
| Pool2 | Average pooling, 2×2, stride 2; output 16×5×5 | 0 |
| Conv3 | 120 filters, 5×5 kernel, Tanh; output 120×1×1 | 48,120 |
| Flatten | Reshape to a 120-dimensional vector | 0 |
| FC1 | Linear 120→84, Tanh; output 84 | 10,164 |
| Output | Linear 84→10 logits, one per digit class | 850 |

The network has 61,706 trainable parameters. It is trained with Adam, batch size 64, learning rate 0.001, weight decay 1e-4, and a step learning-rate scheduler.

## 3 Observations

1. The model classified the clearer digits very confidently, with several predictions close to 100% confidence.
2. Digits 4 and 6 were more difficult because their handwriting styles were less distinct than the simpler digits.
3. The model confused 4 with 8 and 9 with 7, which shows that similar stroke patterns can lead to misclassification.
4. The preprocessing pipeline is important: grayscale conversion, inversion, cropping, padding, resizing, and normalization make custom images more consistent with MNIST.
5. Even with a compact size of 61,706 parameters, LeNet-5 achieved strong performance on handwritten digit recognition, which makes it a good lightweight CNN baseline.

Table 3: Inference results on custom images

| Image | Actual class | Predicted class | Confidence |
|---|---|---|---|
| 0.png | 0 | 0 | 99.99% |
| 1.png | 1 | 1 | 99.99% |
| 2.png | 2 | 2 | 99.89% |
| 3.png | 3 | 3 | 100.00% |
| 4.png | 4 | 8 | 62.25% |
| 5.png | 5 | 5 | 99.98% |
| 6.png | 6 | 6 | 73.87% |
| 7.png | 7 | 7 | 98.86% |
| 8.png | 8 | 8 | 100.00% |
| 9.png | 9 | 7 | 99.95% |

The custom image files are named by their true labels, so 0.png corresponds to class 0, 1.png to class 1, and so on.

The most useful training graphs for this project are the loss curve and the accuracy curve, because they show how quickly the model converged and whether validation performance stayed close to training performance. The plot below is generated from the real 30-epoch training history saved in `lenet5_weights.json`.

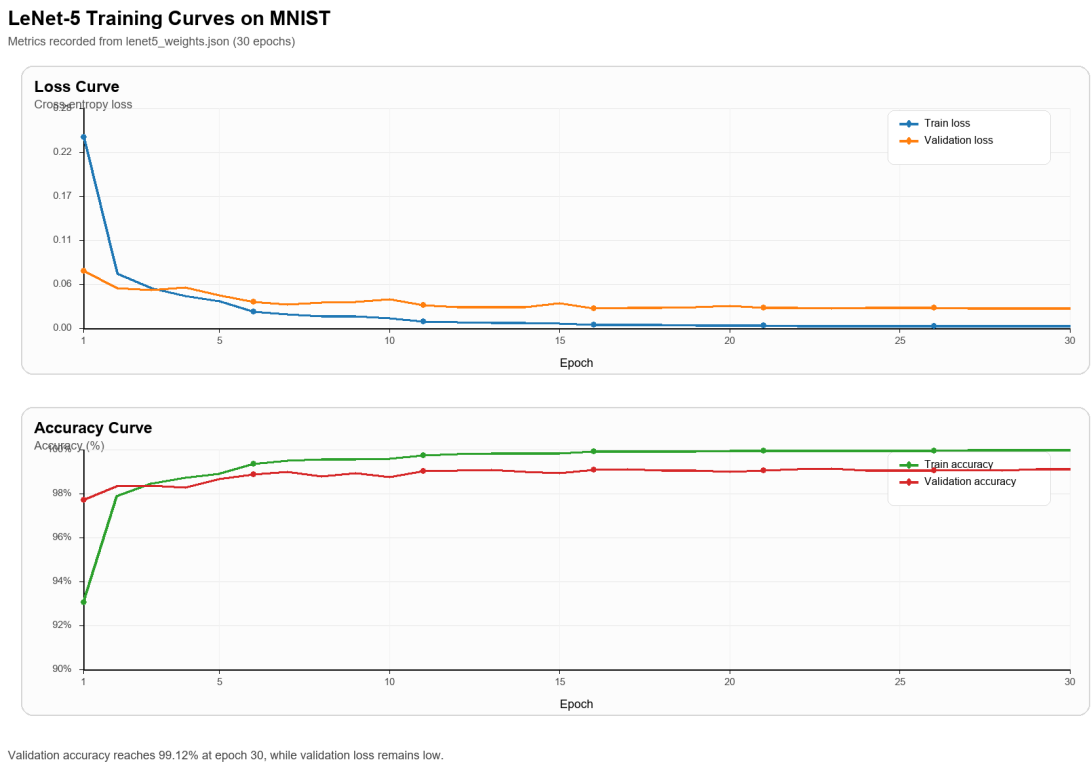


Figure 2: Training and validation loss and accuracy over 30 epochs.

# 4 Conclusion

This project demonstrates that a LeNet-5 style convolutional neural network is effective for MNIST digit recognition. The dataset is simple enough for a compact model, yet it still reveals where ambiguity in handwriting can challenge the classifier. The results show that the model is highly reliable on most digits, while a few visually similar samples remain difficult cases.